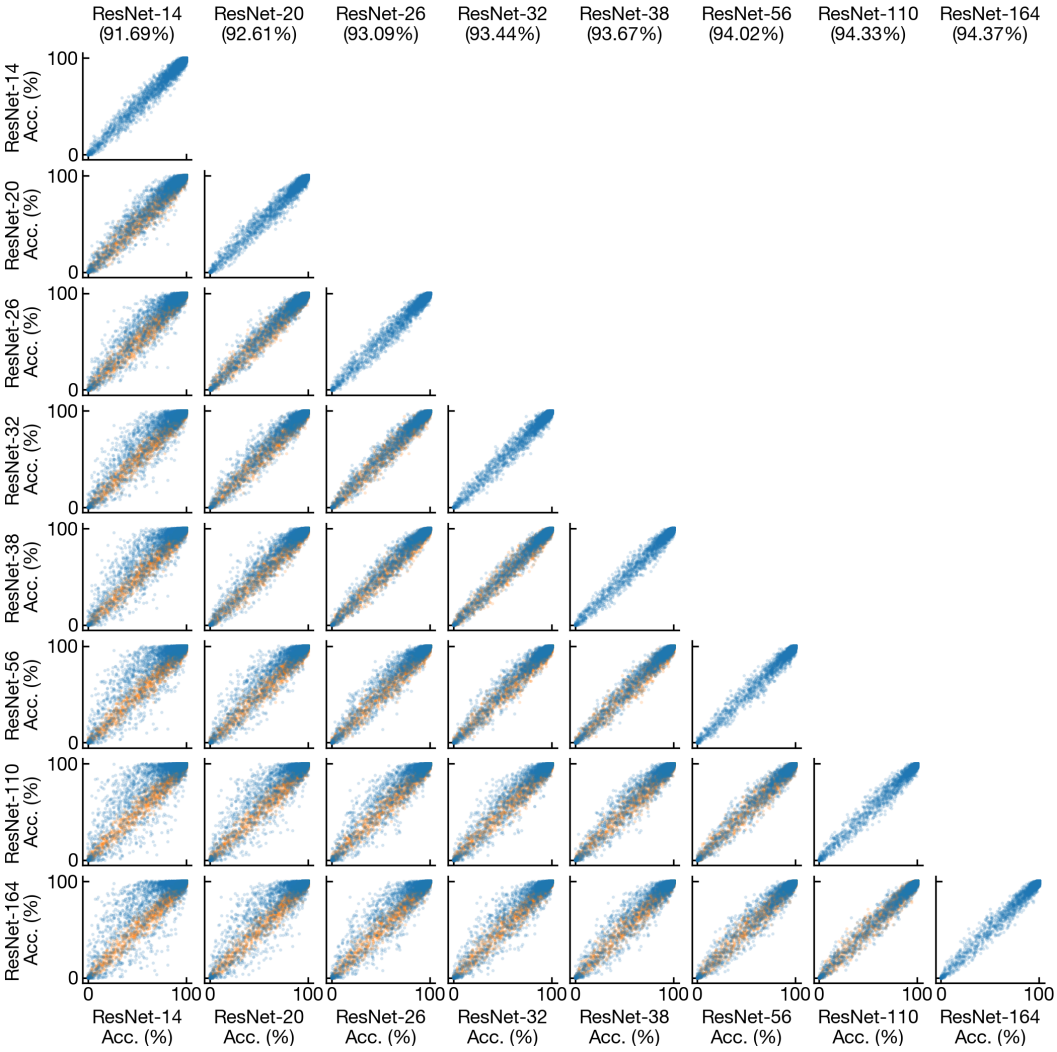

Figure F.2: Effect of depth on example accuracy. Scatter plots of per-example accuracies of ResNet models with different depths on CIFAR-10. Blue dots indicate per-example accuracies of two groups of 100 networks each with different architectures indicated by axes labels. Orange dots show the distribution for groups of architecturally identical models, copied from the plot on the diagonal above. Accuracy of each model is shown at the top.

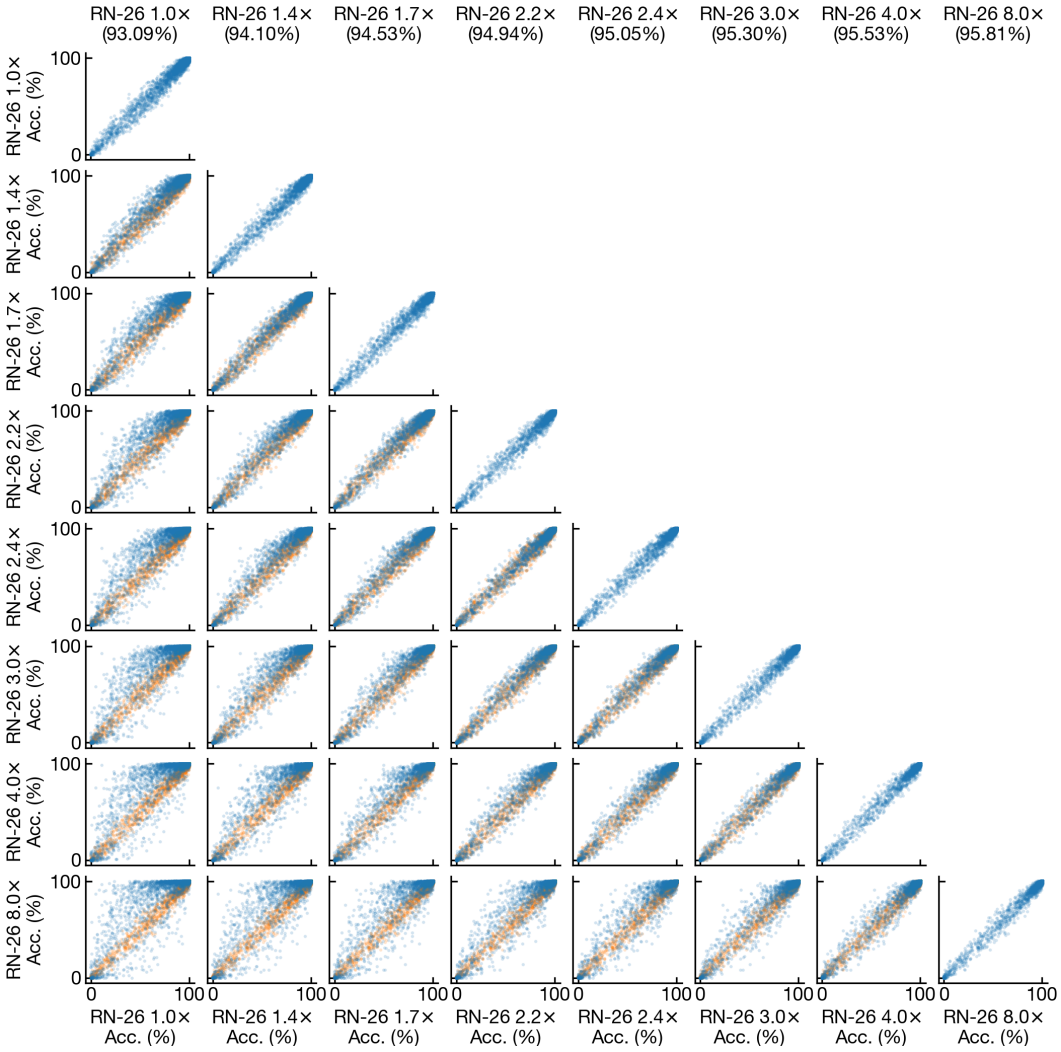

Figure F.3: Effect of width on example accuracy. Scatter plots of per-example accuracies of ResNet models with different widths on CIFAR-10. Blue dots indicate per-example accuracies of two groups of 100 networks each with different architectures indicated by axes labels. Orange dots show the expected distribution for groups of architecturally identical models, copied from the plot on the diagonal above. Accuracy of each model is shown at the top.

F.2 EFFECT OF VARYING WIDTH AND DEPTH ON IMAGENET PREDICTIONS

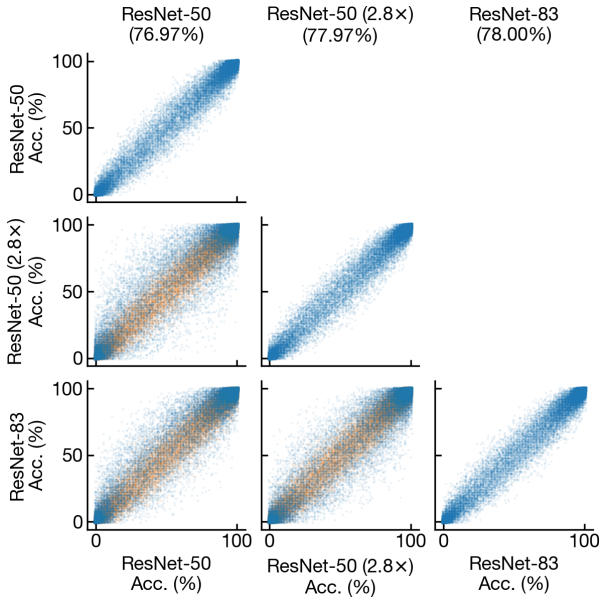

Figure F.4: Systematic per-example performance differences between wide and deep models on ImageNet. Scatter plots of per-example accuracy averaged across 50 vanilla ResNet-50 ($1\times$) models versus that for groups of 50 models with increased depth ($6 \rightarrow 17$ blocks, "ResNet-83") or width ($2.8\times$ wider) in the 3rd stage. Orange dots in plots show the expected distribution for two groups of 50 architecturally identical models, copied from the plot on the diagonal above. The deeper and wider models have very similar but statistically distinguishable accuracy (mean $\pm$ SEM for deeper model: 78.00 ± 0.01, wider model: 77.97 ± 0.01, $t(99) = 2.0$, $p = 0.047$).

F.3 EFFECT SIZES FOR CLASS-LEVEL EFFECTS

We measure how much of the difference between the example-level predictions of the wide and deep ImageNet ResNets in Section 7 could be explained by the classes to which they belonged by fitting a set of three models. Model A attempts to model whether each prediction was correct or incorrect as a linear combination factors corresponding to the example ID and whether the prediction came from a wide or deep model (`statsmodels` formula `y ~ C(example_id) + C(wide_or_deep)`). Model B includes a factor corresponding to the example ID as well as factors corresponding to the interaction between the class ID and the type of model the prediction came from (`statsmodels` formula `y ~ C(example_id) + C(wide_or_deep) * C(class_id)`). Model C includes the interaction between the example ID and the type of model, and corresponds to simply measuring the average accuracy separately for both types of models (`statsmodels` formula `y ~ C(example_id) * C(wide_or_deep)`). Note that model C is nested inside model B, which is nested inside model A.

We seek to measure how much of the variability in predictions that can be explained by model C but not model A can be explained by model B. We fit these models with logistic regression and measure the residual variance $\mathrm{Var}_Q = \sum_{i=1}^{n}(y_i - \pi_i)^2/n$, where n is the number of examples $\times$ the number of models, y_i is either 1 or 0 depending on whether the CNN's prediction was correct or incorrect, and π_i is the output probability from logistic regression model Q. We then compute the squared differences between y and the predictions of the logistic regression model:

$$v^2 = \frac{\mathrm{Var}_A - \mathrm{Var}_B}{\mathrm{Var}_A - \mathrm{Var}_C} = 0.11. \tag{10}$$

This approach is analogous to the pseudo-R^2 of Efron (1978).